

Haoyu Ren

Ph.D. Candidate in Mathematics and Statistics | Biostatistics, Causal Inference, Clinical Trial Methods

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PROFILE

Ph.D. candidate in Mathematics and Statistics with experience in biostatistics, clinical trial methodology, causal inference, statistical modeling, and healthcare data analysis. Experienced in collaborating with clinicians and cross-functional research teams, developing simulation studies, analyzing clinical and real-world healthcare data, and communicating statistical results for oncology, biomarker, and patient outcomes research. Proficient in R, SAS, and statistical programming, with interest in innovative statistical methods for early clinical development and drug/vaccine research.

EDUCATION

University of Maryland, Baltimore County, Baltimore, MD

M.S. in Mathematics and Statistics, 2024

Ph.D. in Mathematics and Statistics, GPA: 3.8/4.0, September 2021 - Present; expected graduation: 2026

Advisors: Dr. Yi Huang and Dr. Chixiang Chen

University of Maryland, College Park, College Park, MD

B.S. in Mathematics and Statistics, cum laude, GPA: 3.9/4.0

EXPERIENCE

Research Assistant | June 2024 - Present

Division of Biostatistics and Bioinformatics, Department of Epidemiology and Public Health, University of Maryland School of Medicine

- Collaborated with clinicians and cross-functional research teams to translate clinical questions into appropriate statistical models, conduct data analyses, and interpret study findings.
- Co-led development of doubly robust and varying-coefficient methods for continuous treatments while addressing missing data.
- Co-led development of a causal meta-analysis weighting function with smaller bias and variance under rare events than existing weighting functions.

Biostatistics Intern | June 2026 - August 2026

Biostatistics and Research Decision Sciences, Merck, Upper Gwynedd, PA

- Evaluated the impact of imaging frequency on progression-free survival (PFS) estimation and hypothesis testing in oncology trials.

Teaching Assistant | September 2021 - May 2024

Department of Statistics, University of Maryland, Baltimore County

- Explained mathematical and statistical concepts to students; held office hours and created and graded quizzes and homework assignments.

SELECTED PUBLICATIONS

- Ren, H., Wen, T., ..., Chen, C. (2026). A Doubly Robust Approach to Evaluating the Impact of Hospital Length of Stay on Post-Fracture Recovery in Older Adults with Cognitive Impairment. Under revision at *Statistics in Medicine*.
- Ren, H., ..., Chen, C. (2026). A Novel Causal Machine Learning Approach to Study Threshold Effects of Continuous Exposure Using Multi-Source Linked Medicare Claims Data. Prepared for submission to *Biostatistics*.
- Shen, B., Ren, H., Shardell, M., Falvey, J., & Chen, C. (2023). Analyzing risk factors for post-acute recovery in older adults with Alzheimer's disease and related dementia: A new semiparametric model for large-scale Medicare claims. *Statistics in Medicine*, 43(5), 1003-1018. <https://doi.org/10.1002/sim.9982>
- Falvey, J. R., Chen, C., ..., Ren, H., ... (2023). Associations of days spent at home before hip fracture with postfracture days at home and 1-year mortality among Medicare beneficiaries living with Alzheimer's disease or related dementias. *The Journals of Gerontology, Series A*, 78(12), 2356-2362. <https://doi.org/10.1093/gerona/glad158>
- Kardon, A., Kim, D., Ren, H., ... (2024). A propensity-score weighted analysis of short-term corticosteroid therapy for refractory pain following spontaneous subarachnoid hemorrhage. *Neurocritical Care*.
- Graeme, F., Chen, C., ..., Ren, H., ... (2025). Microbubble-enhanced focused ultrasound with temozolomide for patients with high-grade gliomas: A multicenter, open-label, phase 1/2 trial. *Lancet Oncology*.
- Kardon, A., Ren, H., ... (2025). Early markers of systemic inflammation are not related to pain burden after aneurysmal subarachnoid hemorrhage: A multicenter observational study. *Neurocritical Care*. DOI: 10.1007/s12028-025-02396-w.
- Blair, R., ..., Ren, H., ... (2026). Post-transcriptional control of HIF-1alpha by MBNL1 restrains hypoxia-driven stemness in GBM. *Neoplasia*. <https://doi.org/10.1016/j.neo.2025.101262>
- Fahim, A., ..., Ren, H., ... (2025). Systemic Cyst(e)inase Administration Induces Ferroptosis and Synergizes with Temozolomide in Glioblastoma. *iScience*. <https://doi.org/10.1016/j.isci.2025.114350>

PRESENTATIONS

- ENAR, New Orleans, March 2025 - contributed paper on heterogeneity of hospital length-of-stay effects on post-acute recovery outcomes for older adults with cognitive conditions.
- JSM, Boston, August 2026 - contributed paper on a causal machine learning approach to threshold effects of continuous exposure.
- ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Maryland, 2026 - poster on a causal machine learning approach to threshold effects of continuous exposure.

SKILLS

Programming and Software: MATLAB, SAS, R, Java, C++, Minitab

Methods and Collaboration: causal inference, statistical modeling, study design, simulation studies, real-world healthcare data analysis, cross-functional collaboration, Microsoft Word, Excel, PowerPoint